

Advanced Data Analysis Course - Pre-Course Refresher Exercises

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Pre-Course Refresher Workshop

Purpose

This refresher is designed to review the most important skills from last year's introductory R course before we begin the 6-week Advanced Data Analysis course. You have already seen most of these tools before. The goal is to rebuild confidence, refresh your memory, and get everyone ready to move into more advanced statistical thinking.

This refresher is divided into two sessions:

- **Week 1: Basics** — project setup, working directories, importing data, exploratory summaries, and basic plots
 - **Week 2: Wrangling** — data wrangling, joins, simple models, transformations, and a short applied challenge
-

Week 1: Basics

0. Setup

Before we begin any coding, we need to make sure your files are organized correctly. Good file organization will save you a lot of confusion later in the course.

0.1. Set up your course folder

For this course, you should keep **all course materials in one main folder** on your computer.

Step 1: Create a main course folder Create a new folder somewhere easy to find on your computer. For example:

```
Advanced_data_analysis_course
```

Step 2: Create a subfolder for this workshop Inside that main folder, create a second folder called: `pre-course refresher`

Your folder structure should look something like this:

```
Advanced_data_analysis_course/  
-pre-course refresher/
```

Step 3: Move your downloaded files You downloaded these two files from Google Drive into your Downloads folder:

- `Radac pre-course refresher.Rmd`
- `penguins.csv`

Move both files from Downloads into the `pre-course refresher` folder.

So now your folder should look like this:

```
Advanced_data_analysis_course/  
-pre-course refresher/  
  -Radac pre-course refresher.Rmd  
  -penguins.csv
```

0.2. Check file names carefully

File names in R must match exactly.

That means:

- uppercase and lowercase letters matter
- spaces matter
- punctuation matters
- file extensions matter

For example:

`penguins.csv` is **not** the same as:

- `Penguins.csv`
- `penguins_.csv`
- `penguins1.CSV`
- `penguins`

Also make sure the file is really a `.csv` file and the R Markdown file is really an `.Rmd` file.

0.3. Create an R Project

We will use an R Project for this course. This is important because it helps RStudio automatically keep track of the correct working directory.

1. Open RStudio
2. Go to: `File > New Project`
3. Choose: `Existing Directory`
4. Navigate to your main course folder:
`Advanced_data_analysis_course`
5. Click `Create Project`

RStudio should now reopen inside your course project.

0.4. Open the refresher file from inside the project

Once the project opens: 1. create a new Markdown file (.Rmd): **File > New File > R Markdown** 2. Name it `wk0_refresher.Rmd`. This will be the script file for the refresher.

0.5. What is a working directory?

Your working directory is the folder R is currently using to look for files and save output.

Because you opened an R Project, your working directory should now be:

```
.../Advanced_data_analysis_course
```

That means your `.csv` file is inside a subfolder called:

```
pre-course refresher
```

Check your working directory

Run:

```
getwd()
```

```
## [1] "/Users/rsteinitz/Documents/github/R Advanced Data Analysis/pre-course refresher"
```

Your output should end with something like:

```
.../Advanced_data_analysis_course
```

If your working directory is NOT the main course folder, set it manually like this:

```
# setwd("~/Advanced_data_analysis_course")
```

Then check again:

```
getwd()
```

```
## [1] "/Users/rsteinitz/Documents/github/R Advanced Data Analysis/pre-course refresher"
```

Check that R can see your files

Run:

```
list.files()
```

```
## [1] "island_info.csv"           "penguins.csv"
## [3] "Project Setup in R - Refresher.pdf" "Radac pre-course refresher.Rmd"
## [5] "Radac-pre-course-refresher_files" "Radac-pre-course-refresher.pdf"
## [7] "Radac-pre-course-refresher.Rmd"
```

You should see:

```
pre-course refresher
```

Now run:

```
list.files("pre-course refresher")
```

```
## character(0)
```

You should see:

```
Radac pre-course refresher.Rmd  
penguins.csv
```

REFRESHER STARTS HERE

1. Load Packages

```
# install the package only if you have not yet, although you should have it already installed;  
# you might want to update by re-installing!  
# -----  
# install.packages(c("tidyverse", "here", "janitor"))  
  
# load packages  
library(tidyverse)  
library(here)  
library(janitor)  
  
# install.packages("skimr")  
library(skimr)
```

2. Import a Dataset

Because your `.csv` file is inside the `pre-course refresher` folder, but your working directory is in `.../Advanced_data_analysis_course`, use this path:

```
penguins <- readr::read_csv(here::here("pre-course refresher/penguins.csv")) %>%  
  janitor::clean_names()
```

This means that from `.../Advanced_data_analysis_course`, you are navigating into the sub-folder `pre-course refresher`, and looking for the file `penguins.csv` there.

3. Warm-Up: Explore the Data

We will use the penguins dataset for most exercises because it contains a mix of numeric and categorical variables and includes some missing values.

3.1. First look

```
dplyr::glimpse(penguins)
```

```
## Rows: 333
## Columns: 19
## $ x1 <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ study_name <chr> "PAL0708", "PAL0708", "PAL0708", "PAL0708", "PAL0708~
## $ sample_number <dbl> 1, 2, 3, 5, 6, 7, 8, 13, 14, 15, 16, 17, 18, 19, 20,~
## $ species <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ region <chr> "Anvers", "Anvers", "Anvers", "Anvers", "Anvers", "A~
## $ island <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ stage <chr> "Adult, 1 Egg Stage", "Adult, 1 Egg Stage", "Adult, ~
## $ individual_id <chr> "N1A1", "N1A2", "N2A1", "N3A1", "N3A2", "N4A1", "N4A~
## $ clutch_completion <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "No", "No", "Yes"~
## $ date_egg <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
## $ flipper_length_mm <dbl> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
## $ body_mass_g <dbl> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~
## $ sex <chr> "MALE", "FEMALE", "FEMALE", "FEMALE", "MALE", "FEMAL~
## $ delta_15_n_o_oo <dbl> NA, 8.94956, 8.36821, 8.76651, 8.66496, 9.18718, 9.4~
## $ delta_13_c_o_oo <dbl> NA, -24.69454, -25.33302, -25.32426, -25.29805, -25.~
## $ comments <chr> "Not enough blood for isotopes.", NA, NA, NA, NA, "N~
## $ date_e <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
```

```
summary(penguins) # summary of the dataset
```

```
##           x1           study_name      sample_number      species
## Min.      : 1   Length:333          Min.      : 1.00   Length:333
## 1st Qu.: 84   Class :character      1st Qu.: 30.00   Class :character
## Median :167   Mode  :character      Median : 58.00   Mode  :character
## Mean    :167                                Mean    : 63.65
## 3rd Qu.:250                                3rd Qu.: 95.00
## Max.    :333                                Max.    :152.00
##
##           region           island           stage           individual_id
## Length:333          Length:333          Length:333          Length:333
## Class :character    Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character    Mode  :character
##
##
##
## clutch_completion    date_egg           bill_length_mm    bill_depth_mm
## Length:333          Length:333          Min.      :32.10    Min.      :13.10
## Class :character    Class :character    1st Qu.:39.50      1st Qu.:15.60
## Mode  :character    Mode  :character    Median :44.50      Median :17.30
##                                Mean    :43.99      Mean    :17.16
##                                3rd Qu.:48.60      3rd Qu.:18.70
##                                Max.    :59.60      Max.    :21.50
##
## flipper_length_mm    body_mass_g           sex           delta_15_n_o_oo
## Min.      :172       Min.      :2700       Length:333       Min.      : 7.632
## 1st Qu.:190       1st Qu.:3550       Class :character    1st Qu.: 8.304
## Median :197       Median :4050       Mode  :character    Median : 8.659
## Mean    :201       Mean    :4207                                Mean    : 8.740
## 3rd Qu.:213       3rd Qu.:4775                                3rd Qu.: 9.181
```

```
## Max.      :231      Max.      :6300      Max.      :10.025
##                                     NA's      :9
## delta_13_c_o_oo      comments      date_e
## Min.      :-27.02      Length:333      Length:333
## 1st Qu.   :-26.33      Class :character      Class :character
## Median    :-25.83      Mode  :character      Mode  :character
## Mean      :-25.68
## 3rd Qu.   :-25.04
## Max.      :-23.79
## NA's      :8
```

```
head(penguins) # shows the first 6 rows
```

```
## # A tibble: 6 x 19
##       x1 study_name sample_number species region island stage individual_id
##   <dbl> <chr>          <dbl> <chr>   <chr> <chr>   <chr>   <chr>
## 1     1 PAL0708             1 Adelie Anvers Torgersen Adult, ~ N1A1
## 2     2 PAL0708             2 Adelie Anvers Torgersen Adult, ~ N1A2
## 3     3 PAL0708             3 Adelie Anvers Torgersen Adult, ~ N2A1
## 4     4 PAL0708             5 Adelie Anvers Torgersen Adult, ~ N3A1
## 5     5 PAL0708             6 Adelie Anvers Torgersen Adult, ~ N3A2
## 6     6 PAL0708             7 Adelie Anvers Torgersen Adult, ~ N4A1
## # i 11 more variables: clutch_completion <chr>, date_egg <chr>,
## #   bill_length_mm <dbl>, bill_depth_mm <dbl>, flipper_length_mm <dbl>,
## #   body_mass_g <dbl>, sex <chr>, delta_15_n_o_oo <dbl>, delta_13_c_o_oo <dbl>,
## #   comments <chr>, date_e <chr>
```

```
tail(penguins) # shows the last 6 rows
```

```
## # A tibble: 6 x 19
##       x1 study_name sample_number species region island stage individual_id
##   <dbl> <chr>          <dbl> <chr>   <chr> <chr>   <chr>   <chr>
## 1   328 PAL0910             63 Chinstrap Anvers Dream Adult, 1~ N98A1
## 2   329 PAL0910             64 Chinstrap Anvers Dream Adult, 1~ N98A2
## 3   330 PAL0910             65 Chinstrap Anvers Dream Adult, 1~ N99A1
## 4   331 PAL0910             66 Chinstrap Anvers Dream Adult, 1~ N99A2
## 5   332 PAL0910             67 Chinstrap Anvers Dream Adult, 1~ N100A1
## 6   333 PAL0910             68 Chinstrap Anvers Dream Adult, 1~ N100A2
## # i 11 more variables: clutch_completion <chr>, date_egg <chr>,
## #   bill_length_mm <dbl>, bill_depth_mm <dbl>, flipper_length_mm <dbl>,
## #   body_mass_g <dbl>, sex <chr>, delta_15_n_o_oo <dbl>, delta_13_c_o_oo <dbl>,
## #   comments <chr>, date_e <chr>
```

```
nrow(penguins) # number of rows, or observations
```

```
## [1] 333
```

```
ncol(penguins) # number of columns, or variables
```

```
## [1] 19
```

Questions

1. How many rows and columns are in the dataset?
2. Which variables are numeric?
3. Which variables are categorical?
4. Do you notice any missing values?

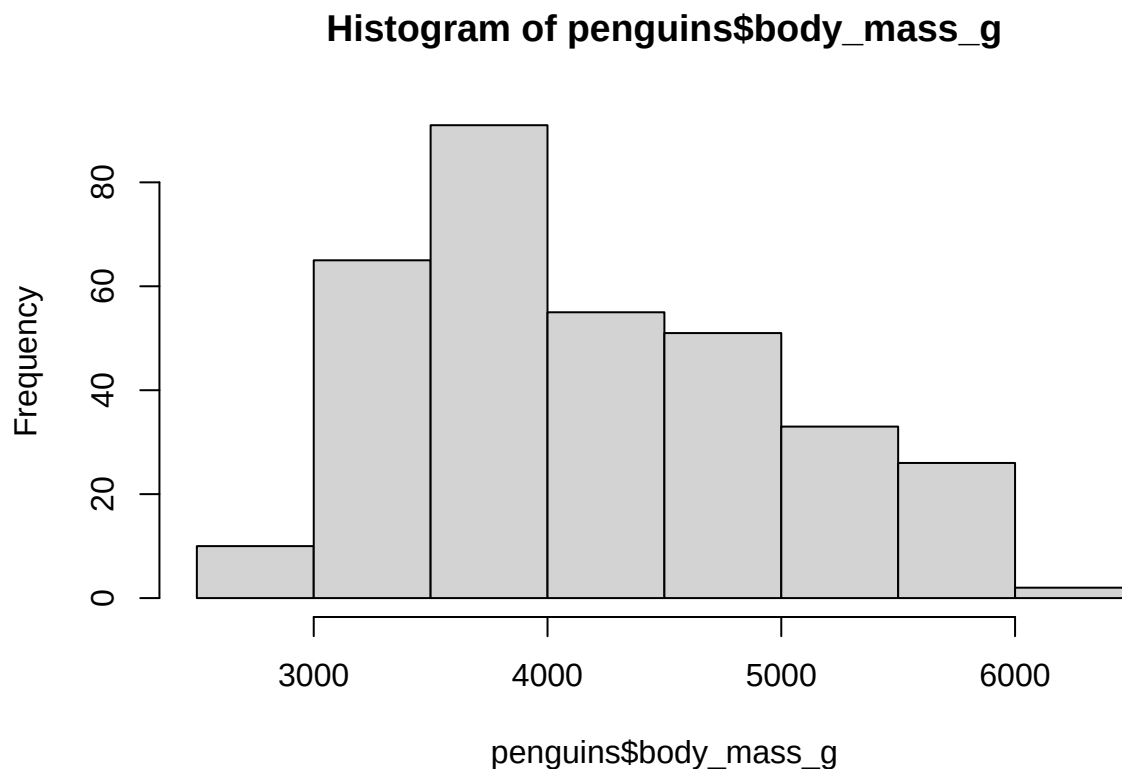
4. Exploratory Plots

Before making polished plots with `ggplot2`, it can be useful to make quick exploratory plots.

4.1. Quick histograms with base R

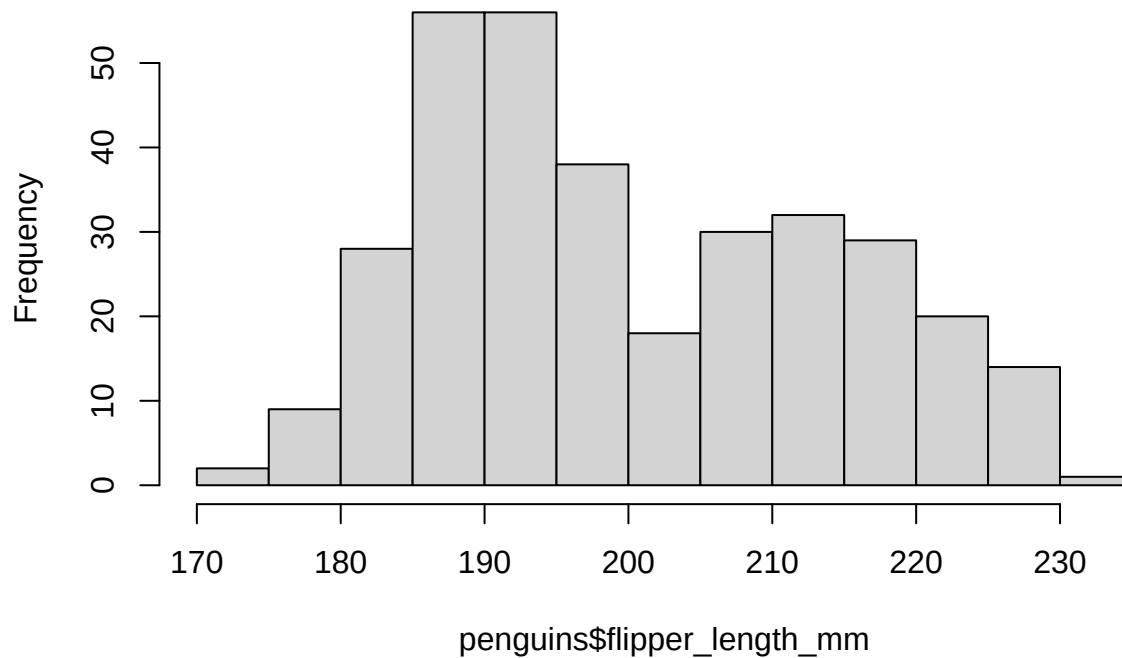
Use `hist()` to look at the distribution of body mass and flipper length.

```
hist(penguins$body_mass_g)
```



```
hist(penguins$flipper_length_mm)
```

Histogram of penguins\$flipper_length_mm



```
# here's a very cool way to combine a numeric summary of the variables with a visual one:  
# skim(penguins)  
### this does not knit well into a pdf, so it's commented out ###
```

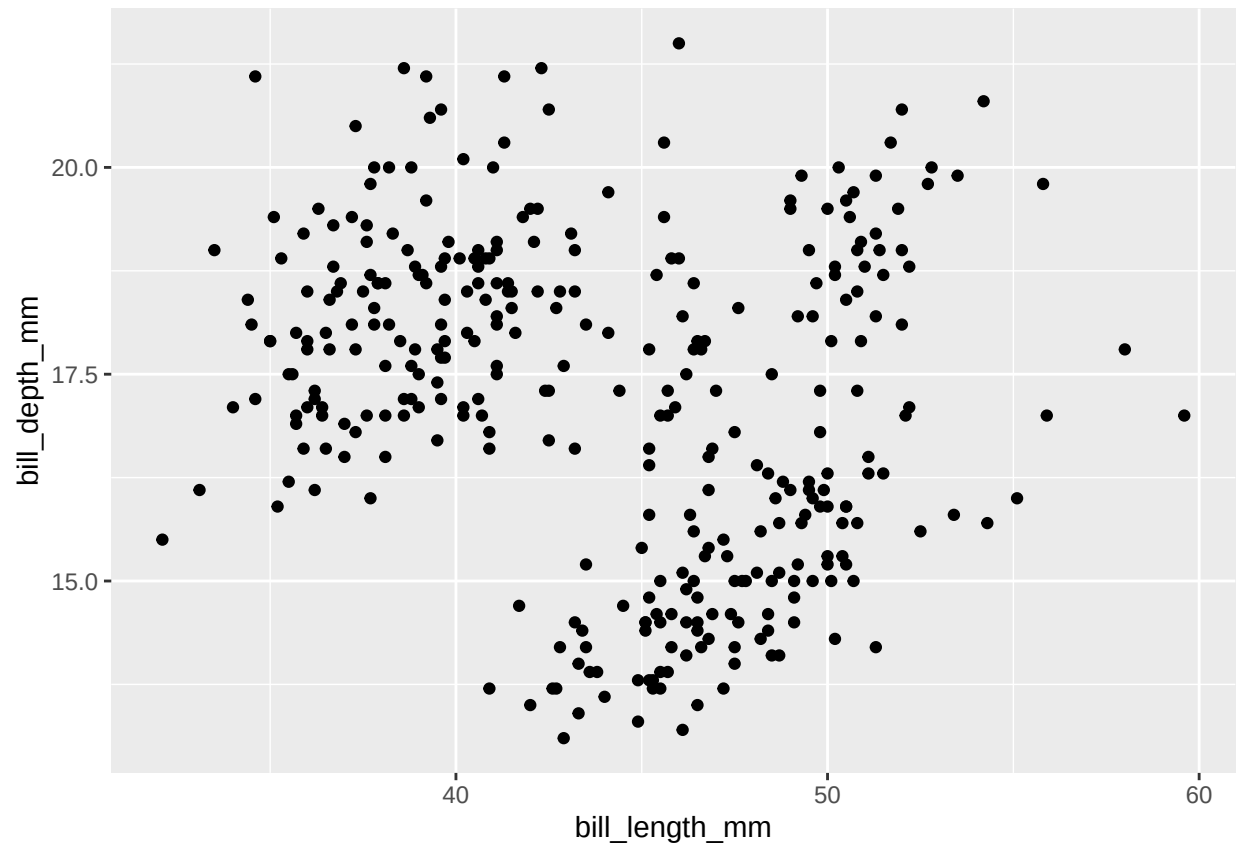
Questions

1. What does each histogram show you?
2. What does the y-axis mean?
3. Do either of these variables look skewed?
4. Are there any values that look unusual?

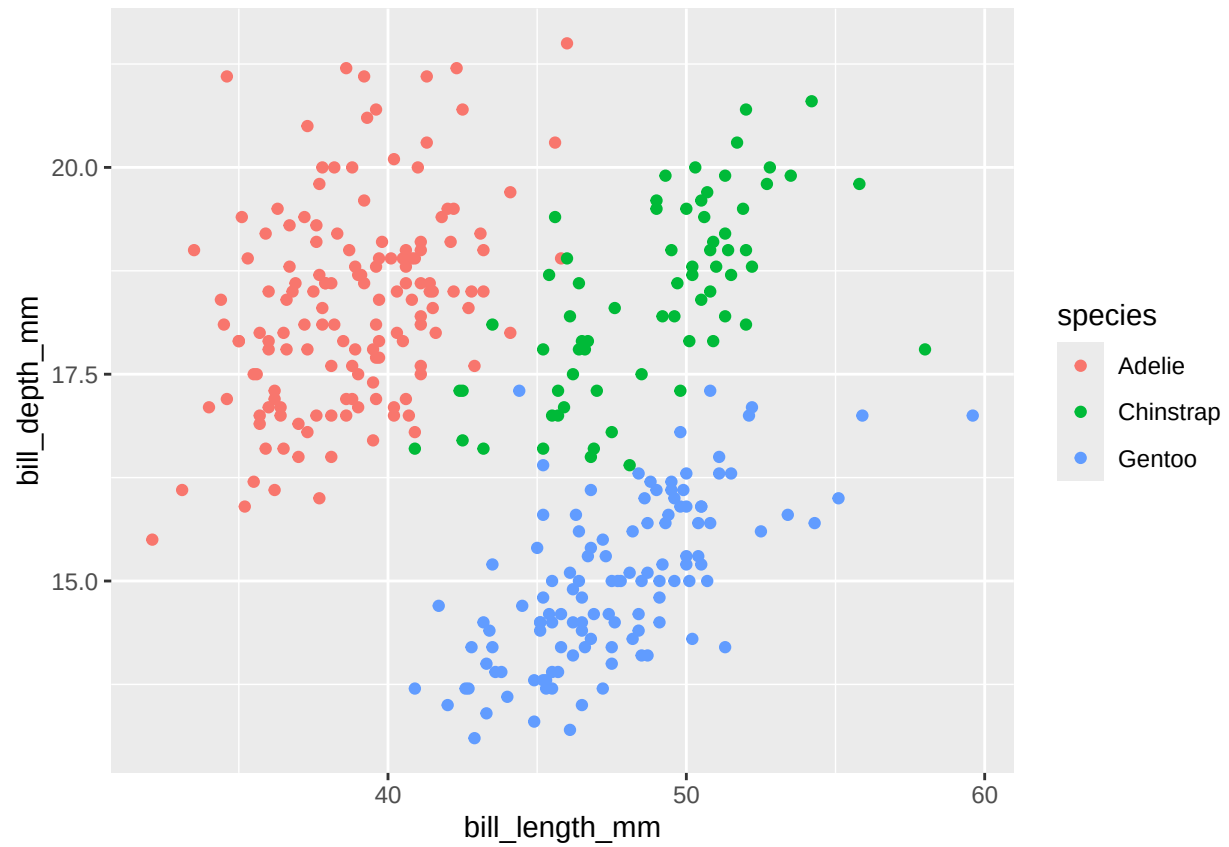
5. Review of Visualization with ggplot2

5.1. Basic visualizations

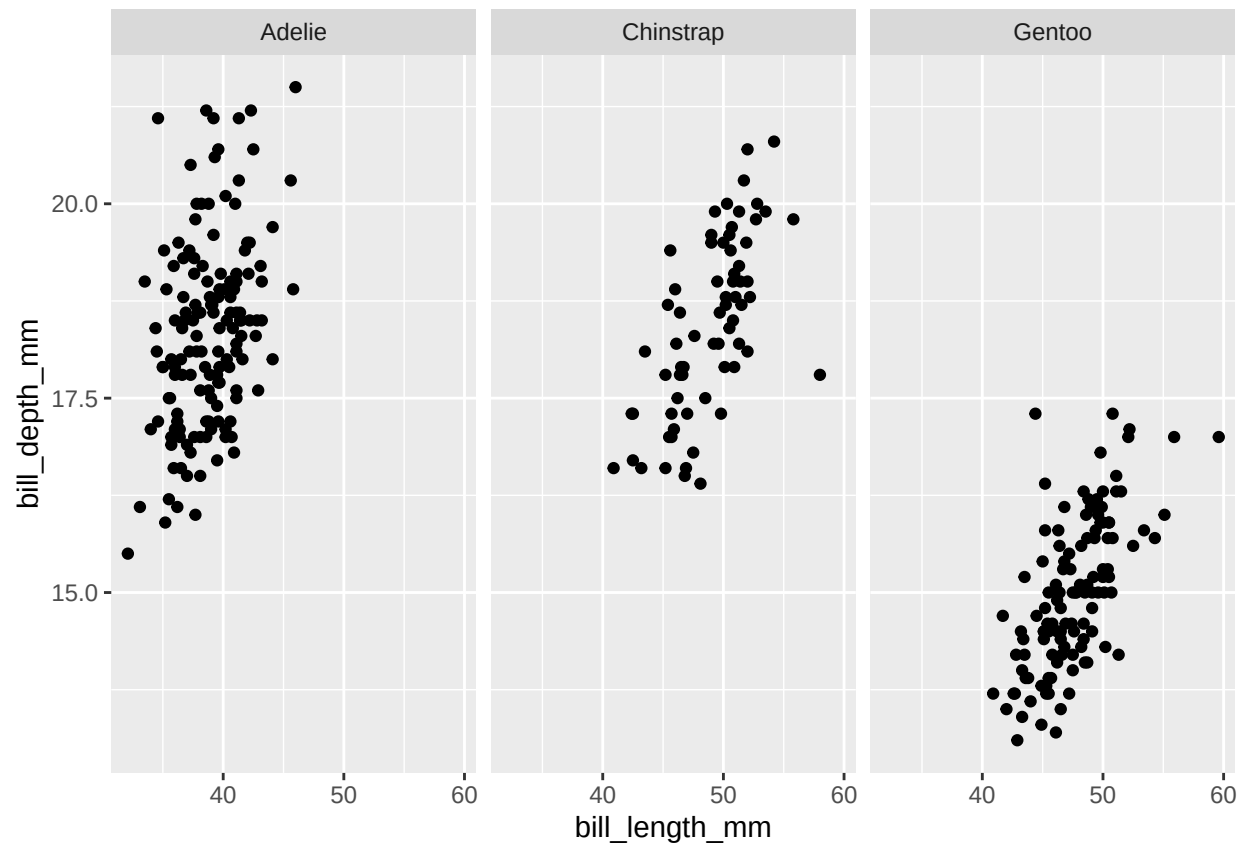
```
# Make a scatterplot of bill length vs. bill depth, colored by species  
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +  
  geom_point()
```

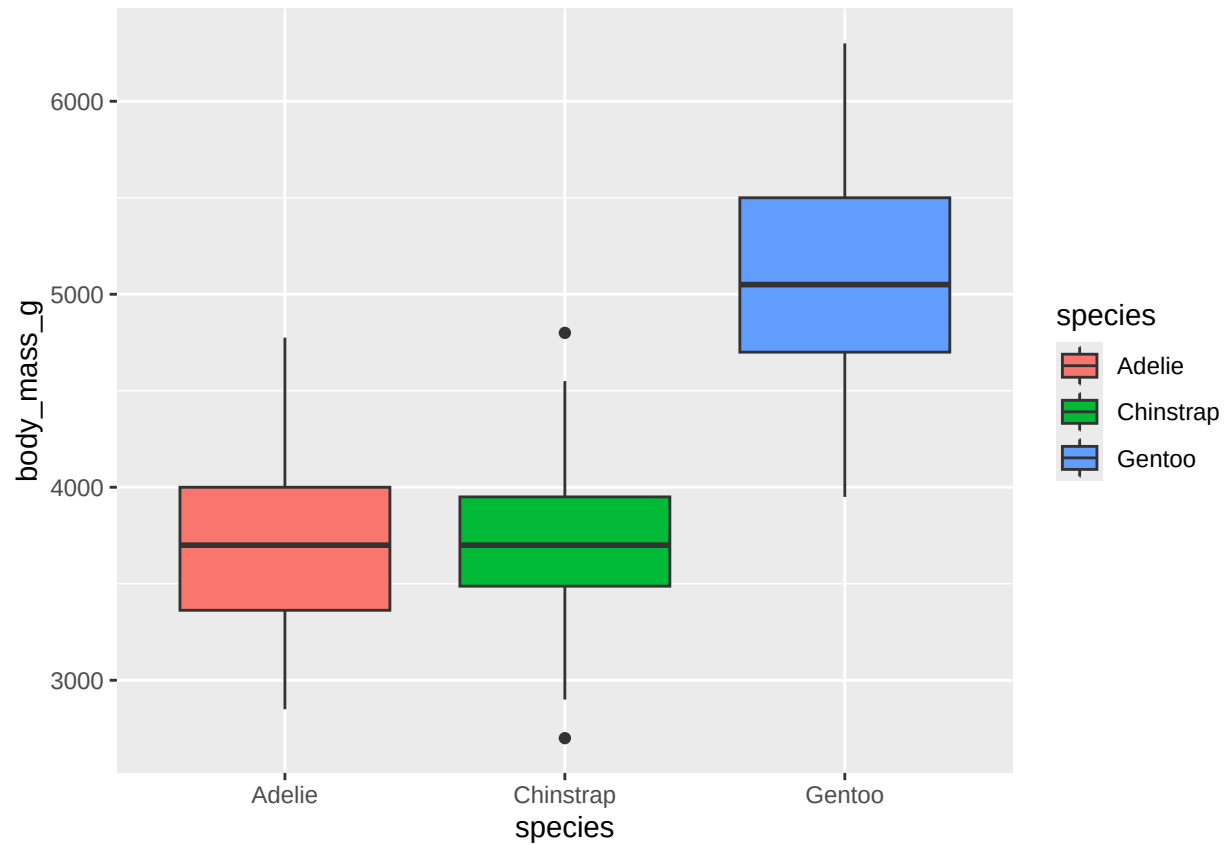
```
# what if I want to see the data for each species?  
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +  
  geom_point()
```



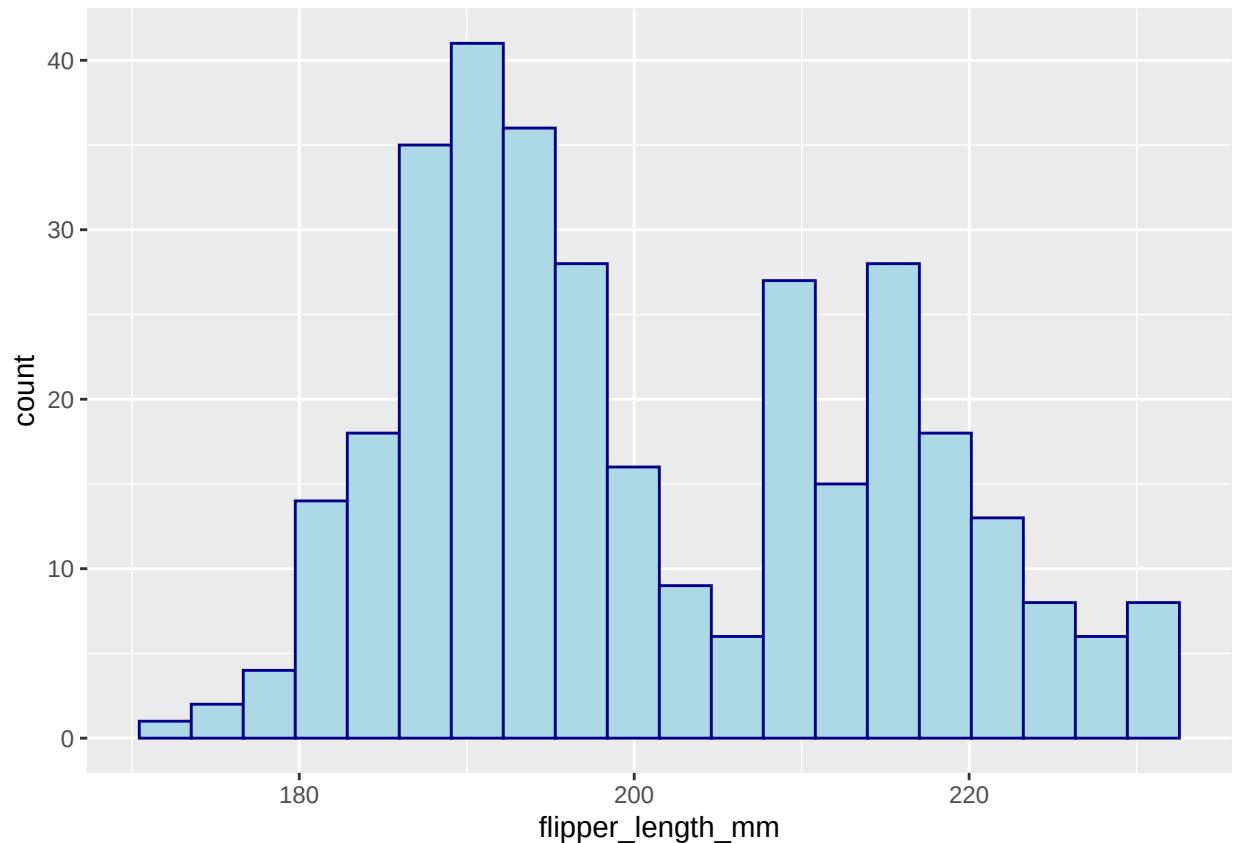
```
# another way of doing this:  
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +  
  geom_point() +  
  facet_wrap(~species)
```



```
# Make a boxplot of body mass by species  
ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +  
  geom_boxplot()
```



```
# Make a histogram of flipper length  
ggplot(penguins, aes(x = flipper_length_mm)) +  
  geom_histogram(bins = 20, fill = "lightblue", color = "darkblue")
```



```
#####
## QUESTION: What does this mean, biologically?
## Explain this graph
#####
```

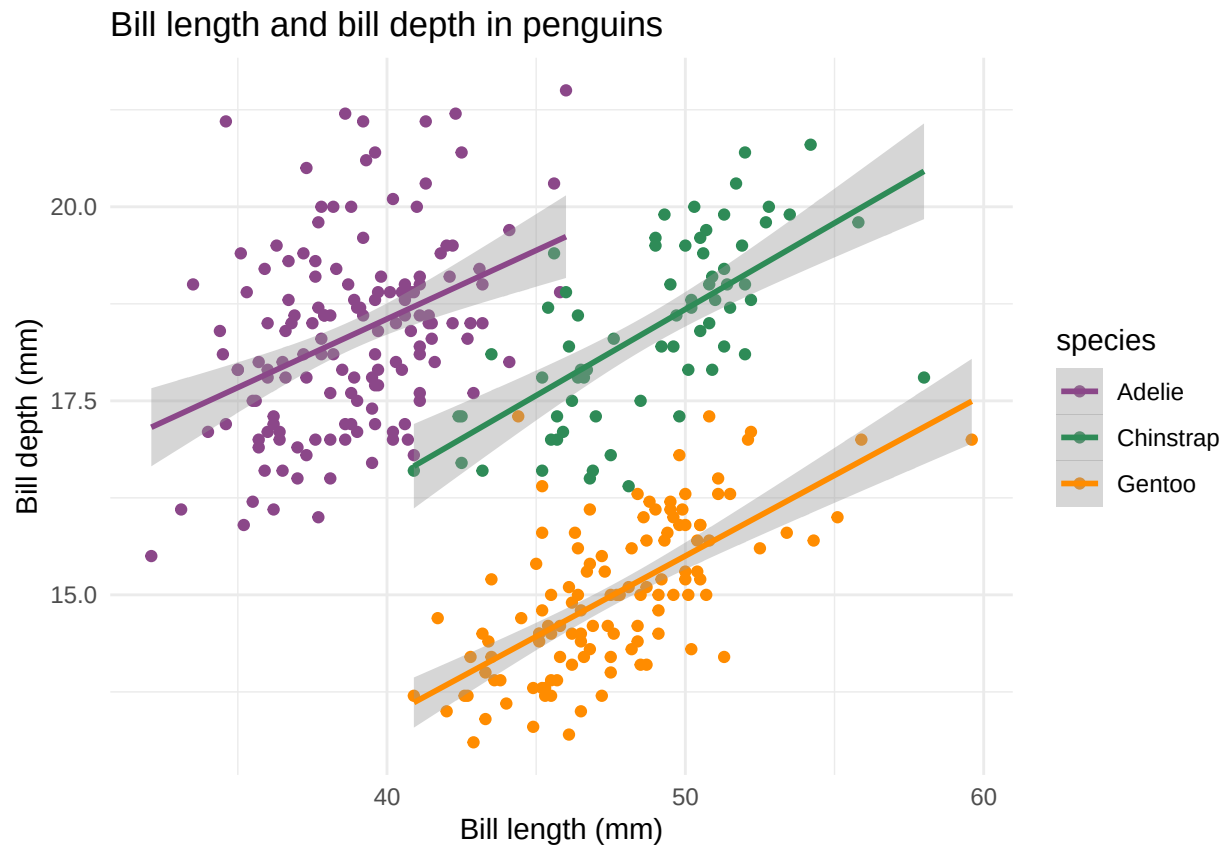
5.2. Improve one plot

Choose one of the plots above and improve it by adding:

- a title
- better axis labels
- custom colors
- a cleaner theme

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Bill length and bill depth in penguins",
    x = "Bill length (mm)",
    y = "Bill depth (mm)"
  ) +
  scale_color_manual(values = c("orchid4", "seagreen", "darkorange")) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Questions

1. Which type of plot is best for comparing groups?
2. Which type is best for showing the relationship between two numeric variables?
3. Which plot would you use to check the distribution of one numeric variable?

Week 2: Wrangling

6. Selecting, Filtering, and Creating Variables

6.1. Select only a few columns

Create a new object called `penguins_small` that includes only these variables:

- `species`
- `island`
- `sex`
- `bill_length_mm`
- `bill_depth_mm`
- `body_mass_g`

```
penguins_small <- penguins %>%
  dplyr::select(species, island, sex, bill_length_mm, bill_depth_mm, body_mass_g)

penguins_small
```

```
## # A tibble: 333 x 6
##   species island    sex  bill_length_mm bill_depth_mm body_mass_g
##   <chr>   <chr>    <chr>        <dbl>         <dbl>      <dbl>
## 1 Adelie Torgersen MALE          39.1          18.7       3750
## 2 Adelie Torgersen FEMALE        39.5          17.4       3800
## 3 Adelie Torgersen FEMALE        40.3          18        3250
## 4 Adelie Torgersen FEMALE        36.7          19.3       3450
## 5 Adelie Torgersen MALE          39.3          20.6       3650
## 6 Adelie Torgersen FEMALE        38.9          17.8       3625
## 7 Adelie Torgersen MALE          39.2          19.6       4675
## 8 Adelie Torgersen FEMALE        41.1          17.6       3200
## 9 Adelie Torgersen MALE          38.6          21.2       3800
## 10 Adelie Torgersen MALE          34.6          21.1       4400
## # i 323 more rows
```

Questions

1. Is this a subset of the original data? Why or why not?
2. Do the means and distributions of the variables change in this new dataset?

6.2. Filter rows

Create a dataset called `female_penguins` that includes only female penguins with non-missing body mass values.

```
female_penguins <- penguins %>%
  dplyr::filter(sex == "FEMALE", !is.na(body_mass_g))

glimpse(female_penguins)
```

```
## Rows: 165
## Columns: 19
## $ x1          <dbl> 2, 3, 4, 6, 8, 11, 12, 14, 16, 18, 21, 23, 24, 26, 2~
## $ study_name  <chr> "PAL0708", "PAL0708", "PAL0708", "PAL0708", "PAL0708~
## $ sample_number <dbl> 2, 3, 5, 7, 13, 16, 17, 19, 21, 23, 26, 28, 29, 31, ~
## $ species     <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ region      <chr> "Anvers", "Anvers", "Anvers", "Anvers", "Anvers", "A~
## $ island      <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ stage       <chr> "Adult, 1 Egg Stage", "Adult, 1 Egg Stage", "Adult, ~
## $ individual_id <chr> "N1A2", "N2A1", "N3A1", "N4A1", "N7A1", "N8A2", "N9A~
## $ clutch_completion <chr> "Yes", "Yes", "Yes", "No", "Yes", "Yes", "Yes", "Yes~
## $ date_egg     <chr> "11/11/07", "11/16/07", "11/16/07", "11/15/07", "11/~
## $ bill_length_mm <dbl> 39.5, 40.3, 36.7, 38.9, 41.1, 36.6, 38.7, 34.4, 37.8~
## $ bill_depth_mm <dbl> 17.4, 18.0, 19.3, 17.8, 17.6, 17.8, 19.0, 18.4, 18.3~
## $ flipper_length_mm <dbl> 186, 195, 193, 181, 182, 185, 195, 184, 174, 189, 18~
```

```
## $ body_mass_g      <dbl> 3800, 3250, 3450, 3625, 3200, 3700, 3450, 3325, 3400~
## $ sex              <chr> "FEMALE", "FEMALE", "FEMALE", "FEMALE", "FEMALE", "F~
## $ delta_15_n_o_oo  <dbl> 8.94956, 8.36821, 8.76651, 9.18718, NA, NA, 9.18528,~
## $ delta_13_c_o_oo  <dbl> -24.69454, -25.33302, -25.32426, -25.21799, NA, NA, ~
## $ comments         <chr> NA, NA, NA, "Nest never observed with full clutch.",~
## $ date_e           <chr> "11/11/07", "11/16/07", "11/16/07", "11/15/07", "11/~
```

Question

1. Is this a subset of the original data? Why or why not?
2. Do the means and distributions of the variables change in this new dataset?
3. How many female penguins are in this filtered dataset?

```
nrow(female_penguins) # you can also look in your "Environment" panel, and see the number of observations
```

```
## [1] 165
```

```
# what proportion of the population is in this dataset?
165/333
```

```
## [1] 0.4954955
```

```
nrow(female_penguins)/nrow(penguins)
```

```
## [1] 0.4954955
```

6.3. Make new variables

Use `dplyr::mutate()` to create:

- `body_mass_kg` = body mass in kilograms
- `bill_ratio` = bill length divided by bill depth

```
penguins_new <- penguins %>%
  dplyr::mutate(
    body_mass_kg = body_mass_g / 1000,
    bill_ratio = bill_length_mm / bill_depth_mm
  )

glimpse(penguins_new)
```

```
## Rows: 333
## Columns: 21
## $ x1              <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ study_name      <chr> "PAL0708", "PAL0708", "PAL0708", "PAL0708", "PAL0708~
## $ sample_number   <dbl> 1, 2, 3, 5, 6, 7, 8, 13, 14, 15, 16, 17, 18, 19, 20,~
## $ species         <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ region          <chr> "Anvers", "Anvers", "Anvers", "Anvers", "Anvers", "A~
```



```
## $ island      <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ stage       <chr> "Adult, 1 Egg Stage", "Adult, 1 Egg Stage", "Adult, ~
## $ individual_id <chr> "N1A1", "N1A2", "N2A1", "N3A1", "N3A2", "N4A1", "N4A~
## $ clutch_completion <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "No", "No", "Yes"~
## $ date_egg     <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
## $ flipper_length_mm <dbl> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
## $ body_mass_g   <dbl> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~
## $ sex           <chr> "MALE", "FEMALE", "FEMALE", "FEMALE", "MALE", "FEMAL~
## $ delta_15_n_o_oo <dbl> NA, 8.94956, 8.36821, 8.76651, 8.66496, 9.18718, 9.4~
## $ delta_13_c_o_oo <dbl> NA, -24.69454, -25.33302, -25.32426, -25.29805, -25.~
## $ comments      <chr> "Not enough blood for isotopes.", NA, NA, NA, NA, "N~
## $ date_e        <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
## $ body_mass_kg   <dbl> 3.750, 3.800, 3.250, 3.450, 3.650, 3.625, 4.675, 3.2~
## $ bill_ratio     <dbl> 2.090909, 2.270115, 2.238889, 1.901554, 1.907767, 2.~
```

Questions

1. Why might `body_mass_kg` be useful?
2. Why might `bill_ratio` be useful?
3. What happens to `bill_ratio` if either bill measurement is missing?

7. Grouping and Summarizing

7.1. Summarize by species

Calculate the following for each species:

- number of penguins
- mean body mass
- mean flipper length

```
species_summary <- penguins %>%
  dplyr::group_by(species) %>%
  dplyr::summarise(
    n = dplyr::n(),
    mean_body_mass = mean(body_mass_g, na.rm = TRUE),
    mean_flipper = mean(flipper_length_mm, na.rm = TRUE)
  )

species_summary
```

```
## # A tibble: 3 x 4
##   species      n mean_body_mass mean_flipper
##   <chr>    <int>      <dbl>      <dbl>
## 1 Adelie    146      3706.        190.
## 2 Chinstrap   68      3733.        196.
## 3 Gentoo    119      5092.        217.
```

```
# if you simply want to see the numbers per species
penguins %>% count(species)
```

```
## # A tibble: 3 x 2
##   species      n
##   <chr>    <int>
## 1 Adelie    146
## 2 Chinstrap  68
## 3 Gentoo    119
```

```
# df %>% count(variable) # for categorical or binary variables
```

7.2. Summarize by species and sex

Now calculate mean body mass by both species and sex.

```
# here we group by more than one category:
species_sex_summary <- penguins %>%
  dplyr::group_by(species, sex) %>%
  dplyr::summarise(
    n = dplyr::n(),
    mean_body_mass = mean(body_mass_g, na.rm = TRUE),
    .groups = "drop"
  )

species_sex_summary
```

```
## # A tibble: 6 x 4
##   species sex      n mean_body_mass
##   <chr>   <chr> <int>         <dbl>
## 1 Adelie FEMALE   73         3369.
## 2 Adelie MALE     73         4043.
## 3 Chinstrap FEMALE  34         3527.
## 4 Chinstrap MALE    34         3939.
## 5 Gentoo  FEMALE   58         4680.
## 6 Gentoo  MALE    61         5485.
```

```
# if you simply want to see the numbers:
penguins %>% count(species, sex)
```

```
## # A tibble: 6 x 3
##   species sex      n
##   <chr>   <chr> <int>
## 1 Adelie FEMALE   73
## 2 Adelie MALE     73
## 3 Chinstrap FEMALE  34
## 4 Chinstrap MALE    34
## 5 Gentoo  FEMALE   58
## 6 Gentoo  MALE    61
```

```
# df %>% count(var1, var2) # again, categorical variables
```

Questions

1. Which species appears heaviest on average?
2. Within species, do males and females differ in body mass?
3. Why is it useful to include `n` when summarizing by group?
-> tells us how many observations were used to calculate the summary statistics for each group. This helps us understand whether a mean is based on many individuals (more reliable) or only a few individuals (less reliable and potentially misleading).

8. Joining Data

8.1. Join the datasets

Import the data and use `dplyr::left_join()` to add the `region` variable to the penguins dataset.

```
island_info <- read_csv(here::here("pre-course refresher/island_info.csv")) %>%
  janitor::clean_names()
```

```
## New names:
## Rows: 3 Columns: 3
## -- Column specification
## ----- Delimiter: "," chr
## (2): island, area dbl (1): ...1
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'
```

```
island_info
```

```
## # A tibble: 3 x 3
##   x1 island area
##   <dbl> <chr> <chr>
## 1     1 Biscoe West
## 2     2 Dream Central
## 3     3 Torgersen East
```

```
penguins_joined <- penguins %>%
  dplyr::left_join(island_info, by = "island") # we joined the "area" information by "island." Each obs
glimpse(penguins_joined)
```

```
## Rows: 333
## Columns: 21
## $ x1.x          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ study_name    <chr> "PAL0708", "PAL0708", "PAL0708", "PAL0708", "PAL0708~
## $ sample_number <dbl> 1, 2, 3, 5, 6, 7, 8, 13, 14, 15, 16, 17, 18, 19, 20,~
```

```
## $ species      <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ region       <chr> "Anvers", "Anvers", "Anvers", "Anvers", "Anvers", "A~
## $ island       <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ stage        <chr> "Adult, 1 Egg Stage", "Adult, 1 Egg Stage", "Adult, ~
## $ individual_id <chr> "N1A1", "N1A2", "N2A1", "N3A1", "N3A2", "N4A1", "N4A~
## $ clutch_completion <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "No", "No", "Yes"~
## $ date_egg      <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
## $ flipper_length_mm <dbl> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
## $ body_mass_g   <dbl> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~
## $ sex           <chr> "MALE", "FEMALE", "FEMALE", "FEMALE", "MALE", "FEMAL~
## $ delta_15_n_o_oo <dbl> NA, 8.94956, 8.36821, 8.76651, 8.66496, 9.18718, 9.4~
## $ delta_13_c_o_oo <dbl> NA, -24.69454, -25.33302, -25.32426, -25.29805, -25.~
## $ comments      <chr> "Not enough blood for isotopes.", NA, NA, NA, NA, "N~
## $ date_e        <chr> "11/11/07", "11/11/07", "11/16/07", "11/16/07", "11/~
## $ x1.y          <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 1, 1, 1~
## $ area          <chr> "East", "East", "East", "East", "East", "East", "Eas~
```

Questions

1. Did the number of rows change after the join? Why or why not?
2. What would happen if one island name were misspelled in one of the datasets?
3. Why do we need a shared column when joining datasets?
 -> *it acts like a common identifier that tells R which rows from the two datasets belong together. Without a shared column (or if the values do not match), R would not know how to correctly combine the information across datasets.*

9. Review of Linear Models

We will fit a simple linear model to test whether flipper length predicts body mass.

9.1. Fit a model

First remove missing values for the two variables.

```
model_data <- penguins %>%
  dplyr::filter(!is.na(flipper_length_mm), !is.na(body_mass_g))
```

Fit the model:

```
mod1 <- lm(body_mass_g ~ flipper_length_mm, data = model_data)

summary(mod1)
```

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = model_data)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1057.33  -259.79   -12.24   242.97  1293.89
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -5872.09     310.29  -18.93  <2e-16 ***
## flipper_length_mm    50.15       1.54   32.56  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 393.3 on 331 degrees of freedom
## Multiple R-squared:  0.7621, Adjusted R-squared:  0.7614
## F-statistic: 1060 on 1 and 331 DF, p-value: < 2.2e-16
```

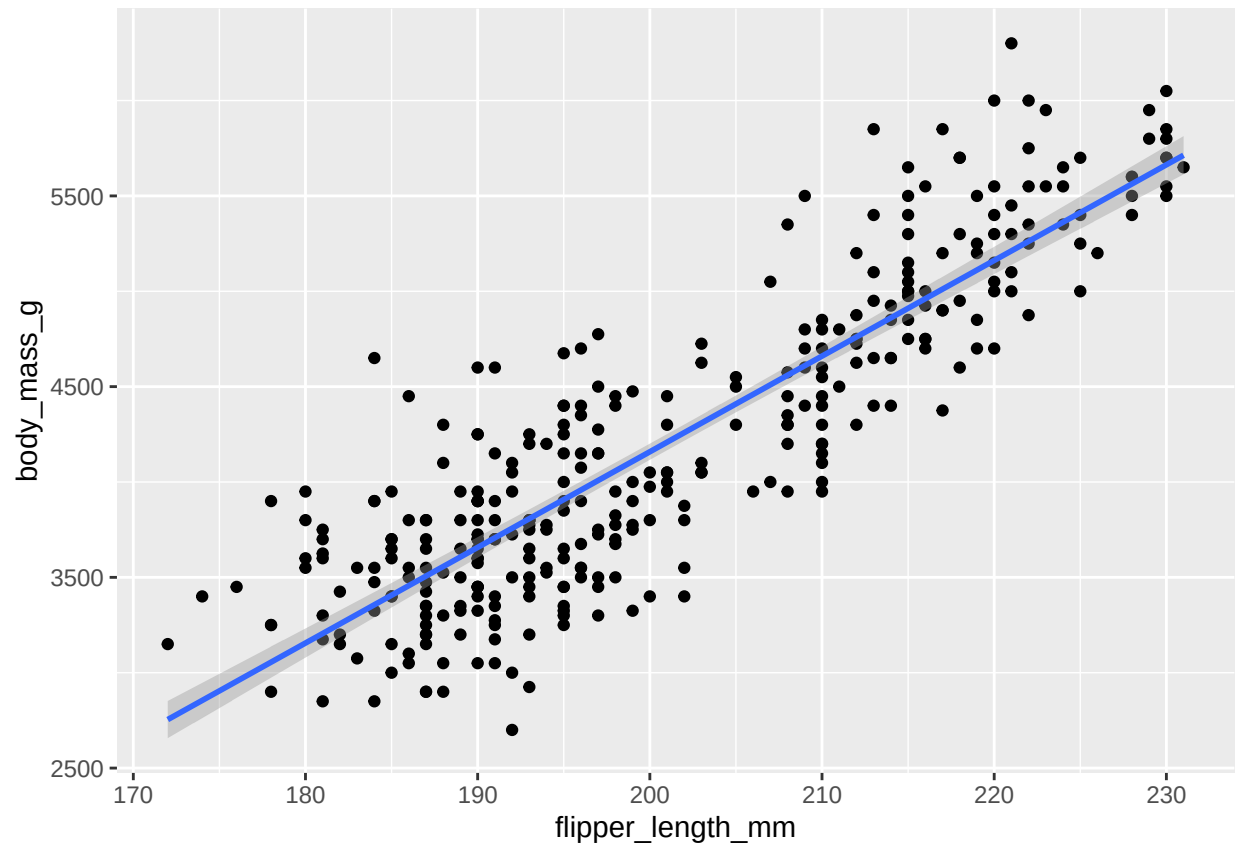
Questions

1. What is the response (outcome) variable in this model? *The response (outcome) variable is body_mass_g because it is the variable being predicted in the model.*
2. What is the predictor variable in this model? *The predictor variable is flipper_length_mm because it is the variable being used to explain or predict body mass.*
3. Is the relationship between flipper length and body mass positive or negative? How can you tell from the output? *The relationship is positive. We can tell because the coefficient (Estimate) for flipper_length_mm is positive (50.15). This means that as flipper length increases, body mass also tends to increase.*
4. What does the intercept represent? Does it make biological sense in this case? *The intercept (-5872.09) represents the predicted body mass when flipper length is 0 mm. This does not make biological sense because a penguin cannot have a flipper length of 0 mm. However, the intercept is still necessary mathematically for fitting the regression line.*
5. What does the p-value tell you about the relationship between flipper length and body mass? *The p-value (< 2e-16) tells us that there is a very strong statistical relationship between flipper length and body mass. The relationship is highly unlikely to be due to random chance alone.*
6. What does the R-squared value tell you? *The R-squared value (0.7621) tells us that about 76% of the variation in penguin body mass can be explained by flipper*

9.2. Visualize the model

```
ggplot2::ggplot(model_data, ggplot2::aes(x = flipper_length_mm, y = body_mass_g)) +
  ggplot2::geom_point() +
  ggplot2::geom_smooth(method = "lm", se = TRUE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Questions

1. Does this visualized relationship match the model output from the linear regression?
2. What does the slope mean in this model, biologically?

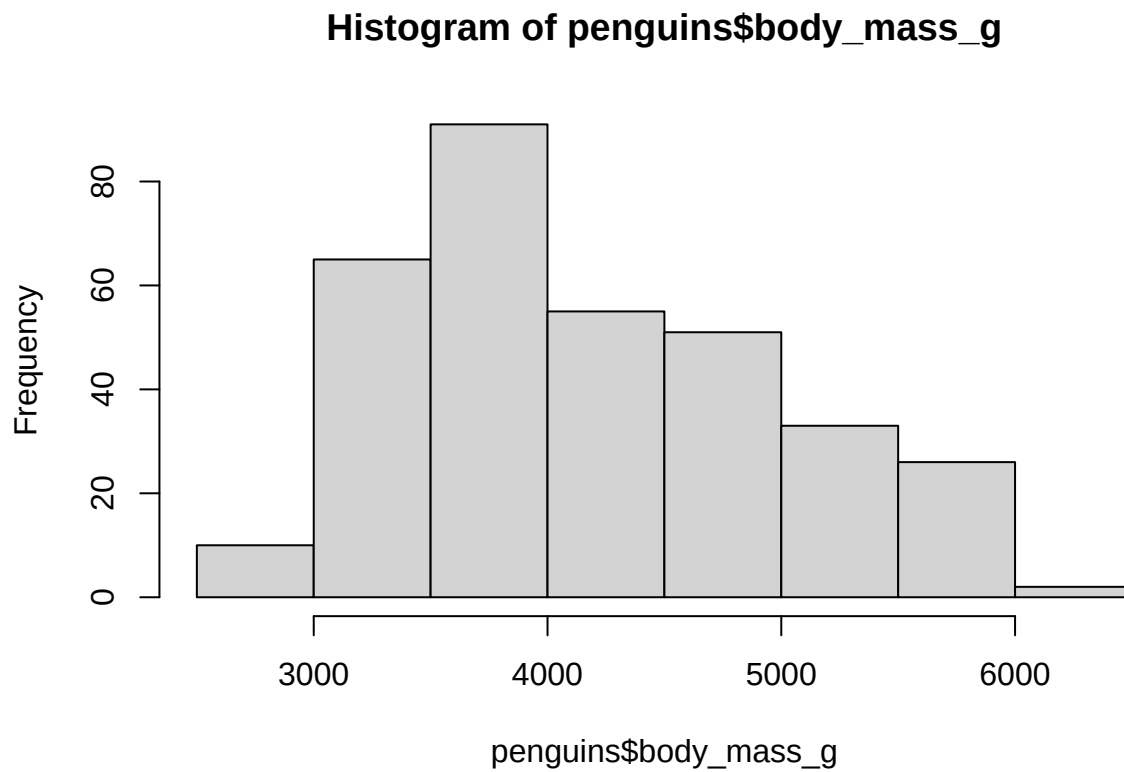
10. Transformations

In the advanced course, we will work more carefully with assumptions and model fit. Transformations are one tool that can help.

10.1. Log-transform a variable

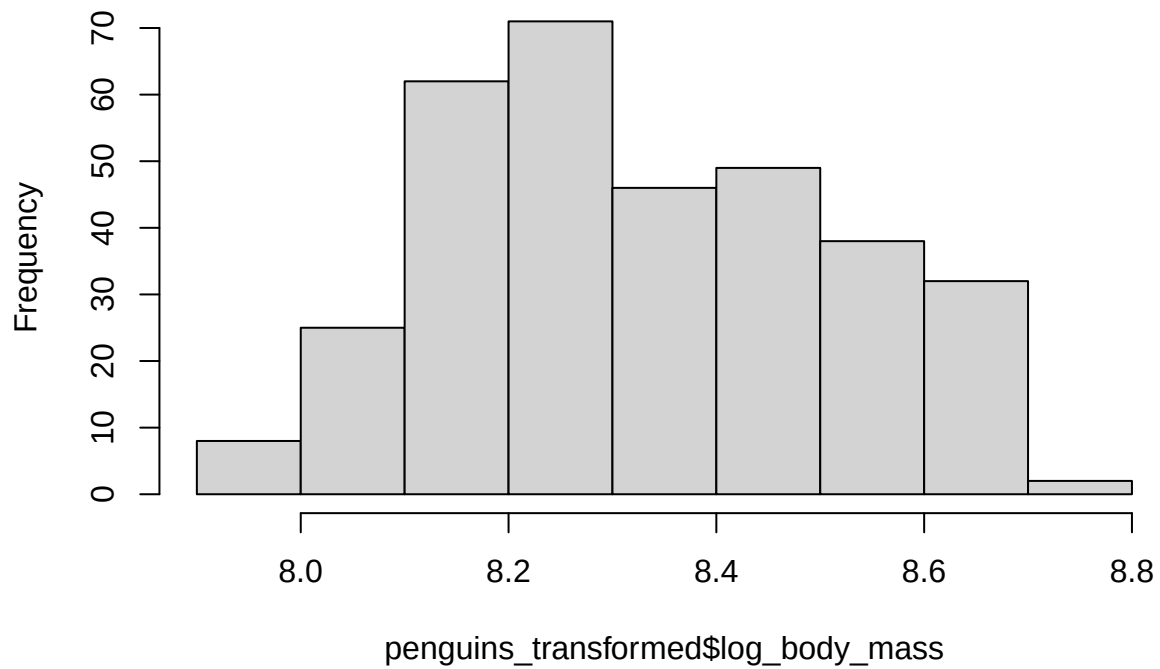
Create a log-transformed version of body mass.

```
hist(penguins$body_mass_g)
```



```
penguins_transformed <- penguins %>%  
  dplyr::filter(!is.na(body_mass_g)) %>%  
  dplyr::mutate(  
    log_body_mass = log(body_mass_g)  
  )  
  
hist(penguins_transformed$log_body_mass)
```

Histogram of penguins_transformed\$log_body_mass

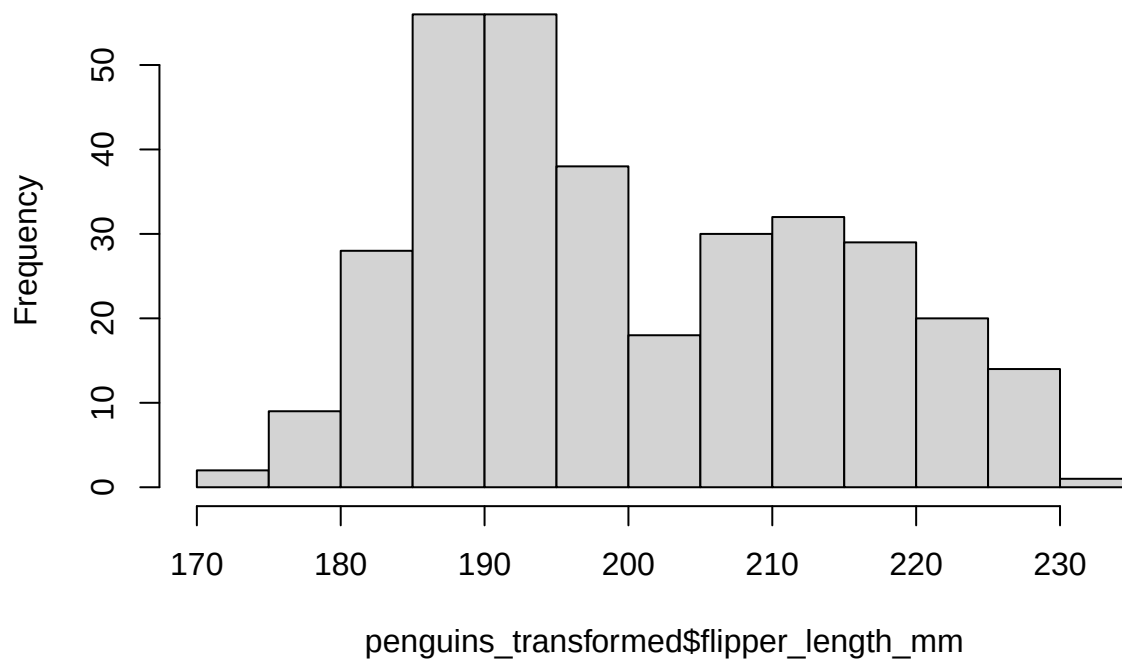


10.2. Standardize a variable

Use `scale()` to standardize flipper length.

```
hist(penguins_transformed$flipper_length_mm)
```


Histogram of penguins_transformed\$flipper_length_mm



```
penguins_transformed <- penguins_transformed %>%  
  dplyr::mutate(  
    flipper_z = as.numeric(scale(flipper_length_mm))  
  )  
  
hist(penguins_transformed$flipper_z)
```

Histogram of penguins_transformed\$flipper_z



Questions

1. What does `log()` do to a skewed variable?
2. What does a z-score represent?
3. Why might standardizing predictors be useful in advanced models?

11. Lab Challenge: Short Applied Practice

Now try a short workflow on your own.

11.1. Mini-analysis

Choose **one** numeric variable and **one** categorical variable from the dataset.

Your task:

1. inspect the variables
2. clean missing values if needed
3. summarize the numeric variable by the categorical variable

4. make one plot
5. write 2-3 sentences interpreting what you found

You can use this template:

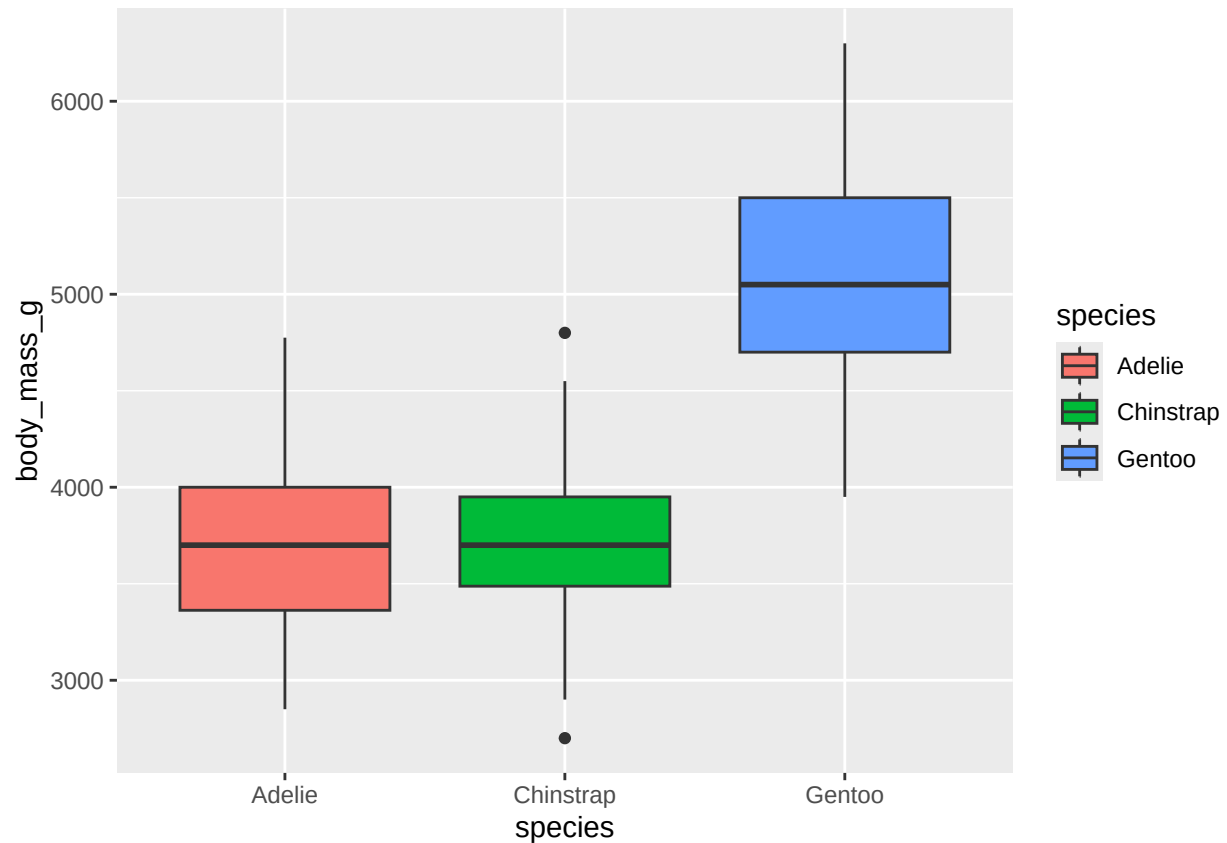
```
# Step 1: choose variables
# Example:
# numeric_var = body_mass_g
# categorical_var = species

mini_summary <- penguins %>%
  dplyr::filter(!is.na(body_mass_g), !is.na(species)) %>%
  dplyr::group_by(species) %>%
  dplyr::summarise(
    mean_value = mean(body_mass_g),
    n = dplyr::n()
  )

mini_summary

## # A tibble: 3 x 3
##   species    mean_value     n
##   <chr>         <dbl> <int>
## 1 Adelie       3706.    146
## 2 Chinstrap   3733.     68
## 3 Gentoo      5092.    119

ggplot2::ggplot(
  penguins %>% dplyr::filter(!is.na(body_mass_g), !is.na(species)),
  ggplot2::aes(x = species, y = body_mass_g, fill = species)
) +
  ggplot2::geom_boxplot()
```



12. Optional Challenge

If you finish early, try one or both of these:

Challenge A:

Fit a linear model with two predictors:

```
mod2 <- lm(body_mass_g ~ flipper_length_mm + species, data = model_data)
```

```
summary(mod2)
```

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + species, data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -898.8  -252.0  -24.8   229.8  1191.6
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -4013.18     586.25  -6.846 3.74e-11 ***
## flipper_length_mm    40.61       3.08  13.186 < 2e-16 ***
```

```
## speciesChinstrap  -205.38      57.57   -3.568 0.000414 ***
## speciesGentoo     284.52      95.43    2.981 0.003083 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 373.3 on 329 degrees of freedom
## Multiple R-squared:  0.787, Adjusted R-squared:  0.7851
## F-statistic: 405.3 on 3 and 329 DF,  p-value: < 2.2e-16
```

How does this model differ from the first one?

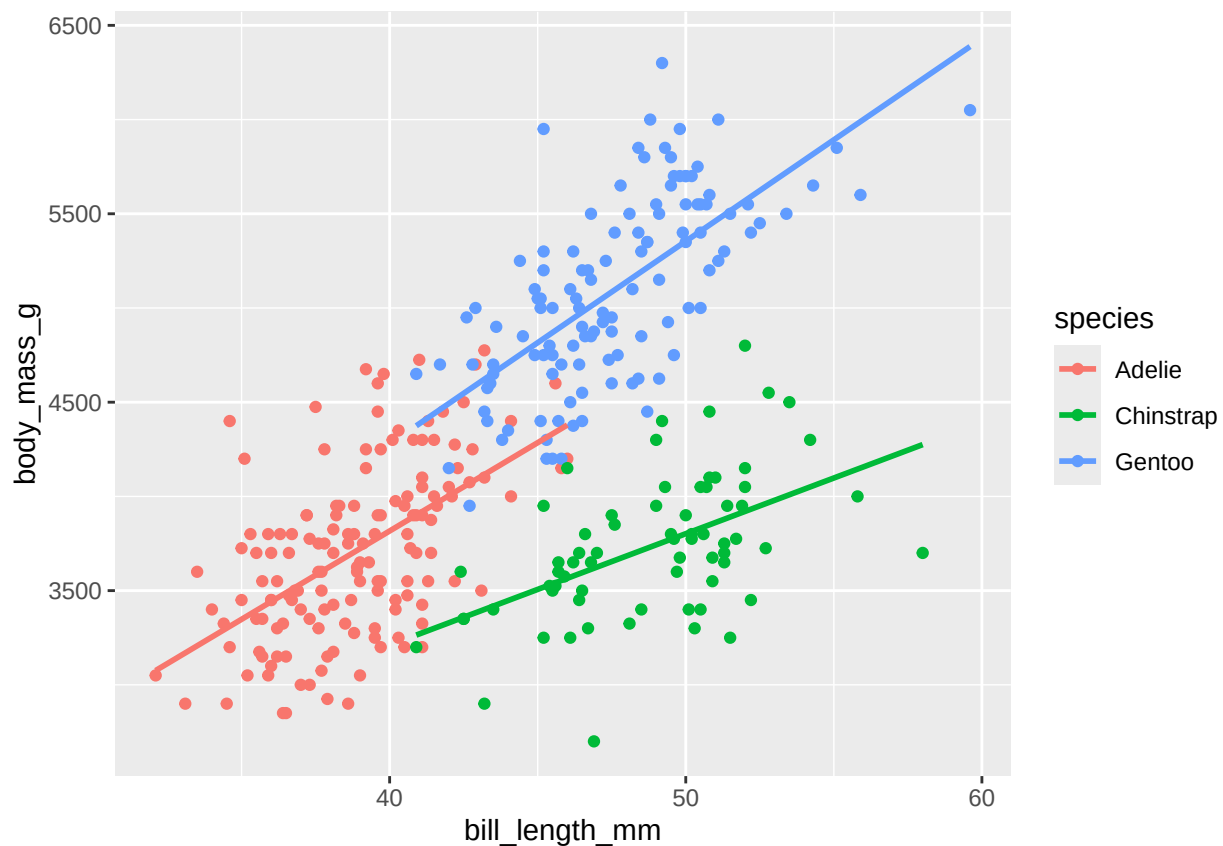
Challenge B:

Make a scatterplot of bill length vs. body mass, colored by species, and add separate trend lines for each species.

```
penguins2 <- penguins %>% filter(!is.na(bill_length_mm) & !is.na(body_mass_g) & !is.na(species))
# removing any observations with missing data

ggplot2::ggplot(penguins2, aes(x = bill_length_mm, y = body_mass_g, color = species)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



What do the species-specific patterns look like?

Use your own dataset for a short refresher analysis.

Choose **one variable** from your own dataset and do the following:

1. clean or filter the data if needed
2. make one plot
3. write 3-4 sentences interpreting the variable
4. if possible, compare it across one grouping variable, such as sex, age class, location, group, treatment, or year

Final Reflection

Before the advanced course begins, make sure you can answer these questions:

- Can I set up and open an R Project?
- Can I check my working directory?
- Can I import a dataset and inspect its structure?
- Can I make quick exploratory summaries and plots?
- Can I filter rows and create new variables?
- Can I summarize values by group?
- Can I join two datasets correctly?
- Can I make and customize a basic plot?
- Can I fit and interpret a simple linear model?
- Can I explain why I used a transformation?

If not, revisit the relevant section and practice again.